

File permissions in Linux

Project description

For this lab scenario, I reviewed and corrected file and directory permissions for a research team in the /home/researcher2/projects directory. I used Linux commands to inspect the current authorization settings, identify permissions that were too open, and update them with chmod. The goal was to make sure access matched the organization's policy and to reduce unnecessary permissions on project files.

Check file and directory details

I first moved into the projects directory and listed all of its contents with detailed permissions. I used ls with the -l option for a long listing and the -a option to include hidden files. This was important because one of the files, .project_x.txt, is hidden and would not appear in a basic ls output.

```
cd /home/researcher2/projects
```

```
ls -la
```

```
researcher2@cc13a61862f9:~$ cd /home/researcher2/projects
researcher2@cc13a61862f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 11 22:51 .
drwxr-xr-x 3 researcher2 research_team 4096 May 11 23:23 ..
-rw--w---- 1 researcher2 research_team  46 May 11 22:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 11 22:51 drafts
-rw-rw-rw- 1 researcher2 research_team  46 May 11 22:51 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May 11 22:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_t.txt
researcher2@cc13a61862f9:~/projects$
```

The output showed one hidden file, one subdirectory, and four visible project files. The first column in each row is the 10-character permissions string. I used that string to decide which permissions needed to be changed.

Describe the permissions string

The permissions string is a compact way to show the file type and the access granted to the user, group, and other users. I used project_t.txt as an example because its permissions were clear in the first listing:

```
-rw-rw-r-- project_t.txt
```

The first character shows the item type. A hyphen (-) means a regular file, while d means a directory. The next nine characters are split into three groups of three: user, group, and other. In each group, r means read, w means write, x means execute, and a hyphen means that permission is not granted.

- The first character, '-', shows that project_t.txt is a regular file.
- The user permissions are rw-, so the owner can read and write the file but cannot execute it.
- The group permissions are rw-, so members of research_team can read and write the file but cannot execute it.
- The other permissions are r--, so all other users can only read the file.

This same pattern applies to every row in the `ls -la` output. For directories, execute permission has a special role because it allows a user to enter or traverse the directory.

Change file permissions

The organization does not allow other users to have write access to project files. In the first listing, `project_k.txt` had permissions of `-rw-rw-rw-`, which meant other users could read and write the file. To fix that, I removed write permission from other with `chmod o-w`.

```
chmod o-w project_k.txt
```

```
ls -la
```

```
researcher2@cc13a61862f9:~/projects$ chmod o-w project_k.txt
researcher2@cc13a61862f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 11 22:51 .
drwxr-xr-x 3 researcher2 research_team 4096 May 11 23:23 ..
-rw--w---- 1 researcher2 research_team  46 May 11 22:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 11 22:51 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May 11 22:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_t.txt
```

After the command, `project_k.txt` changed from `-rw-rw-rw-` to `-rw-rw-r--`. The user and group still had read/write access, but other users were left with read-only access. This matched the requirement without removing permissions that were still authorized.

Change file permissions on a hidden file

The file `.project_x.txt` is hidden because its name starts with a period. Since the project was archived, nobody should have write access to it. The user and group should still be able to read it, and other users should have no access.

```
chmod u-w,g-w,g+r .project_x.txt
```

```
ls -la
```

```
researcher2@cc13a61862f9:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@cc13a61862f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 11 22:51 .
drwxr-xr-x 3 researcher2 research_team 4096 May 11 23:23 ..
-r--r----- 1 researcher2 research_team  46 May 11 22:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 11 22:51 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May 11 22:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_t.txt
```

This `chmod` command made three changes at once. The `u-w` part removed write permission from the user. The `g-w` part removed write permission from the group. The `g+r` part added read permission for the group. After the change, `.project_x.txt` showed `-r--r-----`, which means the user and group can read the file, but no one can write to it and other users have no permissions.

Change directory permissions

The drafts directory should only be accessible by the researcher2 user. In the first listing, the directory permissions were drwx--x---, so the group still had execute permission. On a directory, execute permission allows traversal, so I removed it from the group.

chmod g-x drafts

ls -la

```
researcher2@cc13a61862f9:~/projects$ chmod g-x drafts
researcher2@cc13a61862f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 11 22:51 .
drwxr-xr-x 3 researcher2 research_team 4096 May 11 23:23 ..
-r--r----- 1 researcher2 research_team  46 May 11 22:51 .project_x.txt
drwx----- 2 researcher2 research_team 4096 May 11 22:51 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May 11 22:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 11 22:51 project_t.txt
```

After the command, drafts changed to drwx-----. The researcher2 user kept read, write, and execute permissions, while the group and other users had no access. I did not need to add any user permissions because the user already had full access to the directory.

Summary

I used ls -la to review the permissions in the projects directory, including hidden files, and chmod to remove unauthorized access from project_k.txt, .project_x.txt, and drafts. The final permissions match the scenario and apply least privilege.